Phonon-Drag-Augmented Piecewise Johnson–Cook Modeling for Cold-Spray Impact

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1 Introduction

Cold-spray particle impact involves large plastic deformation, severe strain-rate amplification near the contact interface, and rapid conversion of plastic work into heat [4, 6]. In this regime, a constitutive law must reproduce at least four coupled effects: strain hardening, low-to-moderate strain-rate strengthening, thermal softening, and an additional high-rate resistance once the plastic strain rate exceeds a transition scale. A conventional Johnson–Cook (JC) law provides a practical starting point for the first three of these effects, but its standard logarithmic rate term is generally too weak to reproduce the sharp rise of flow stress observed in the extreme-rate regime.

Recent impact and indentation studies reinforce this point. High-velocity microparticle impact experiments and simulations show that rebound, flattening, jetting, and adhesion do not evolve smoothly under a single weak rate-sensitive law; instead, they exhibit a marked transition as the impact severity increases [7, 8]. In addition, indentation-based ultra-high-strain-rate reconstructions indicate that representative flow strength rises strongly in the range from about 10^5 to 10^6 s^{-1} and above, which is not well captured by the standard JC logarithmic rate term alone [3, 2]. These observations provide the practical constitutive motivation for introducing a separate high-rate strengthening branch.

The motivation for the present work is therefore twofold. First, a low-rate JC branch is retained because it remains a compact and interpretable engineering description of rate- and temperature-dependent plastic flow. Second, a separate high-rate branch is introduced to represent the additional drag-dominant strengthening associated with fast dislocation motion in a thermally activated crystal lattice. This additional branch is not intended as a full dislocation-dynamics model. Instead, it is a physics-informed continuum closure designed for impact simulations in which a tractable constitutive form is needed, but where the transition to drag-controlled behavior must still be tied to physically interpretable scales such as the Burgers vector, the shear modulus, and a characteristic transverse sound speed [1, 5].

A central design choice in this work is that the effective transition strain rate is treated as a material-dependent constitutive threshold, not as a direct function of the externally prescribed impact velocity. The impact velocity determines the actually attained strain-rate field in the particle and substrate, whereas the constitutive transition rate determines when the drag-dominant strengthening mechanism begins to contribute. This distinction proved important during model development and is preserved explicitly in the final formulation.

The paper is organized as follows. Section 2 introduces the phonon-drag-motivated constitutive closure and its coupling to the JC baseline and the Taylor–Quinney heat-generation term. Section 3 summarizes the numerical implementation used in the peridynamic impact solver. The emphasis throughout is on preserving a transparent link between the continuum constitutive form and the material scales used to interpret the transition to the high-rate regime.

2 Method

2.1 Material-Dependent Phonon-Drag Closure

The constitutive form is written in piecewise form because the literature surveyed for cold-spray impact does not support a single weak logarithmic rate sensitivity over the full loading range. High-velocity microparticle impact studies indicate a distinct transition from ordinary plastic flow to a regime characterized by rapid strengthening, jetting, strong interface plasticity, and a marked change in rebound or adhesion behavior as the impact severity increases [3, 7, 8, 2]. This is consistent with the model-development conclusion reached earlier in the present work: the low-rate Johnson–Cook branch should be retained where its logarithmic rate term remains adequate, while an additional drag-controlled increment must be introduced once the local plastic strain rate exceeds a transition threshold. The total flow stress is therefore written in the following piecewise form. At the highest level, the constitutive structure may be summarized as

$$\sigma_{\text{flow}} = \begin{cases} \sigma_{\text{JC,low}}, & \dot{\varepsilon}_p \leq \dot{\varepsilon}_c^{\text{eff}}, \\ \sigma_{\text{JC,low}} + \sigma_{\text{drag}}, & \dot{\varepsilon}_p > \dot{\varepsilon}_c^{\text{eff}}, \end{cases} \tag{1}$$

This summary equation is expanded below. For plastic strain rates below the transition threshold, the standard low-rate Johnson–Cook branch is retained [4],

$$\sigma_{\text{flow}} = \left(A + B\varepsilon_p^n\right)\left[1 + C_1 \ln\left(\frac{\dot{\varepsilon}_p}{\dot{\varepsilon}_0}\right)\right]\left(1 - T^{*m_{\text{th}}}\right), \qquad \dot{\varepsilon}_p \leq \dot{\varepsilon}_c^{\text{eff}}, \tag{2}$$

where

$$T^* = \text{clamp}\left(\frac{T - T_{\text{ref}}}{T_m - T_{\text{ref}}}, 0, 1\right). \tag{3}$$

In Eq. (2), A, B, and n control the hardening response, C_1 controls the low-rate logarithmic strain-rate sensitivity, and m_{th} controls thermal softening. This same low-rate JC contribution is retained as the baseline term even after the transition is crossed.

For plastic strain rates above the transition threshold, the low-rate Johnson–Cook branch is continued and the drag contribution is added as a continuous increment,

$$\sigma_{\text{flow}} = \left(A + B\varepsilon_p^n\right)\left[1 + C_1 \ln\left(\frac{\dot{\varepsilon}_p}{\dot{\varepsilon}_0}\right)\right]\left(1 - T^{*m_{\text{th}}}\right) + \sigma_{\text{drag}}, \qquad \dot{\varepsilon}_p > \dot{\varepsilon}_c^{\text{eff}}. \tag{4}$$

This was the final model decision in the present development: the high-rate branch should not consist of drag alone, but neither is it necessary to freeze the low-rate JC term once the drag increment becomes dominant. Instead, the low-rate JC branch remains as the baseline stress and the drag term contributes only the additional high-rate increment. Because Eq. (21) is zero at $\dot{\varepsilon}_c^{\text{eff}}$, the piecewise law remains continuous at the transition point. The limiting case $K_0 = 0$ therefore reduces to the continued low-rate JC branch, which is the correct zero-increment limit of the present closure.

2.2 Phonon-Drag-Controlled Rate Effect

The thermal phonons in a crystal are scattered by gliding dislocations, thereby resulting in a drag force on the dislocations [1]. By analogy with linear drag, the dislocation drag force per unit length may be written as

$$F = Bv, \tag{5}$$

where B is a drag coefficient and v is the dislocation velocity. Equation (5) is used here as the physical starting point, not as the final constitutive law. The present objective is to construct a continuum closure for the drag-controlled high-rate resistance that appears in Eq. (4).

Following the total–particular–total logic adopted during model development, the drag contribution is first introduced in the generic macroscopic form

$$\sigma_{\mathrm{drag}} = K_0 \times (\text{rate term}), \tag{6}$$

where K_0 is the baseline magnitude of the macroscopic drag resistance. In the present logarithmic closure, K_0 carries units of stress and directly sets the amplitude of the drag-induced stress increment. The remaining task is therefore to determine the temperature scaling and the over-threshold rate measure.

The temperature-dependent amplitude is written in the separable form

$$K_0^{\mathrm{mat}} = K_0 \left(\frac{G(T)}{G_0}\right)^a, \tag{7}$$

where G_0 is the reference shear modulus and a is a temperature-sensitivity exponent. As discussed previously, the characteristic transverse sound speed is treated as a constant,

$$c_T = c_{T0} = \text{constant}, \tag{8}$$

while the temperature dependence is retained only in the shear modulus,

$$G(T) = G_0 \left[1 - \beta_G \frac{T - T_0}{T_m - T_0}\right]. \tag{9}$$

Hence, c_{T0} supplies the velocity scale for high-rate dislocation motion, whereas $G(T)$ controls the temperature-dependent amplitude of the drag resistance.

The macroscopic plastic strain rate is linked to dislocation motion through the Orowan picture, for which

$$\dot{\varepsilon}_p \sim \rho_m b v, \tag{10}$$

where ρ_m is the mobile dislocation density and b is the Burgers vector [5]. In the present continuum closure, the detailed dislocation-density evolution is not resolved explicitly. Instead, Eq. (10) motivates the use of $\dot{\varepsilon}_p$, b, and c_{T0} to define the transition-rate scale.

Blaschke et al. [1] showed that the anharmonic interaction and scattering of phonons by a moving dislocation, namely the phonon wind, imparts a drag force $vB(v, T, \rho)$ on the dislocation. Their analysis further shows that the drag coefficient is only approximately velocity-independent in the low-velocity regime, whereas it increases nonlinearly as the dislocation velocity approaches the transverse sound speed c_T. In that sense, the sound-speed scale provides the physically motivated upper-bound kinematic reference for drag-dominant strengthening. If Eq. (10) is evaluated at the sonic-limit velocity scale, the corresponding microscopic rate scale is first written in the more complete form

$$\dot{\varepsilon}_{\mathrm{limit}} \sim \rho_m b c_{T0}. \tag{11}$$

Equation (11) is the direct Orowan-based estimate. However, the present continuum formulation does not evolve the detailed mobile dislocation density, and the unresolved density scale is absorbed into the bridge calibration introduced below. At the order-of-magnitude level, this corresponds to replacing the unresolved microscopic density factor by a characteristic inverse area scale, $\rho_m \sim O(b^{-2})$, so that

$$\rho_m b c_{T0} \sim \frac{c_{T0}}{b}. \tag{12}$$

Accordingly, the microscopic transition-rate scale used in the present closure is written as

$$\dot{\varepsilon}_c^{\text{micro}} \sim \frac{c_{T0}}{b}. \tag{13}$$

Equation (13) should therefore be interpreted as a recast order-of-magnitude estimate derived from Eq. (11), rather than as a direct one-step substitution from dislocation velocity to strain rate. At the continuum level, the experimentally relevant transition occurs at a much smaller rate. An effective critical strain rate is therefore introduced first,

$$\dot{\varepsilon}_c^{\text{eff}} = \frac{1}{\Lambda_{\text{ph}}} \frac{c_{T0}}{b}, \tag{14}$$

and the bridge factor Λ_{ph} is then specified below. This ordering is important: the constitutive law is defined first in terms of the effective transition rate, and the bridge factor is subsequently used to connect that transition rate to the microscopic sonic-limit estimate.

The critical strain rate is treated here as a material-dependent transition threshold rather than a loading-condition-dependent quantity. The impact velocity controls the actually attained plastic strain-rate field, but it should not directly redefine the intrinsic transition rate itself. Accordingly, $\dot{\varepsilon}_c^{\text{eff}}$ is taken as a fixed constitutive parameter once the material and discretization scales are selected.

To ensure that the high-rate branch is continuous at the transition point without producing an excessively steep high-rate tail, the rate amplification is written in logarithmic form,

$$A_{\dot{\varepsilon}}(\dot{\varepsilon}_p) = \left[\ln \left(\frac{\dot{\varepsilon}_p}{\dot{\varepsilon}_c^{\text{eff}}} \right) \right]_+, \tag{15}$$

where $[x]_+ = \max(x, 0)$. The drag increment therefore vanishes exactly at the transition rate and then grows only logarithmically above it. Combining Eqs. (6), (7), and (15) yields the working drag closure

$$\sigma_{\text{drag}} = K_0 \left(\frac{G(T)}{G_0} \right)^a \left[\ln \left(\frac{\dot{\varepsilon}_p}{\dot{\varepsilon}_c^{\text{eff}}} \right) \right]_+, \tag{16}$$

In Eq. (16), the logarithmic over-threshold term controls the shape of the high-rate rise. This was adopted specifically to avoid the overly slow mid-rate growth and overly steep ultra-high-rate growth produced by the earlier multiplicative rate-power form.

The bridge factor is introduced to connect the continuum transition rate to the microscopic scale in Eq. (13). A simple linear scale ratio d/b is too small to reduce the transition rate to the experimentally relevant range, while a volume-type ratio $(d/b)^3$ over-suppresses the transition rate. In the present formulation, an area-type bridge is adopted,

$$\Lambda_{\text{ph}} = \left(\frac{d}{b} \right)^2, \tag{17}$$

where d is the continuum discretization length. For the current implementation,

$$d = 1.0e - 6 \text{ m}, \qquad b = 2.56e - 10 \text{ m}, \tag{18}$$

which gives

$$\Lambda_{\mathrm{ph}} \approx 1.53 \times 10^{7}, \tag{19}$$

and therefore

$$\dot{\varepsilon}_c^{\mathrm{eff}} \approx 5.7 \times 10^{5}\ s^{-1}. \tag{20}$$

Substituting Eq. (14) into Eq. (16) finally gives the fully substituted drag law,

$$\sigma_{\mathrm{drag}} = K_0 \left(\frac{G(T)}{G_0}\right)^{a} \left[\ln\left(\Lambda_{\mathrm{ph}} \frac{\dot{\varepsilon}_p b}{c_{T0}}\right)\right]_{+}, \tag{21}$$

Equation (21) is the final closed form used in the constitutive model.

2.3 Taylor–Quinney and Johnson–Cook Coupling

The thermomechanical coupling is introduced through the Taylor–Quinney relation [6, 9]. The adiabatic temperature rise associated with plastic work is written as

$$\Delta T = \frac{\beta_{\mathrm{TQ}}\, \sigma_{\mathrm{flow}}\, \Delta\varepsilon_p}{\rho c_p}, \tag{22}$$

where β_{TQ} is the Taylor–Quinney coefficient, ρ is the density, and c_p is the specific heat. Equation (22) feeds back into the constitutive model through two paths: the thermal softening factor in Eq. (2) and the modulus-ratio term in Eq. (21).

Table 1 summarizes the meanings of the main model parameters. As requested during model writing, the table is restricted to parameter meaning only; the role of each parameter in the constitutive law is described directly below the corresponding equations rather than being duplicated in the table.

Table 1: Meaning of the main model parameters.

Symbol	Meaning
A	Reference yield stress
B	Strain-hardening coefficient
n	Strain-hardening exponent
C_1	Low-rate logarithmic strain-rate coefficient
$\dot{\varepsilon}_0$	Reference plastic strain rate
m_{th}	Thermal softening exponent
β_{TQ}	Taylor–Quinney coefficient
K_0	Baseline macroscopic drag stress scale
a	Exponent for drag-amplitude temperature sensitivity
Λ_{ph}	Macro-to-micro bridge factor
b	Burgers vector
c_{T0}	Characteristic transverse sound speed
G_0	Reference shear modulus
β_G	Shear-modulus thermal degradation coefficient
d	Continuum discretization length used in the bridge factor

Table 2 lists representative material scales for Cu and Al that were discussed during model construction.

Table 2: Representative material scales used to interpret the drag transition.

Material	ρ (kg/m^3)	b (nm)	G_0 (GPa)	c_{T0} (m/s)	c_{T0}/b (s^{-1})
Cu	8960	0.256	44.8	2.24×10^3	8.75×10^{12}
Al	2700	0.286	26.3	3.12×10^3	1.09×10^{13}

3 Numerical Implementation

3.1 Johnson–Cook Stress Evaluation

The reconstructed constitutive model is implemented through the Johnson–Cook force-model path in the impact solver. At each material point, the constitutive update requires the accumulated plastic strain ε_p, the instantaneous plastic strain rate $\dot{\varepsilon}_p$, and the adiabatic temperature rise. The low-rate branch evaluates the standard JC expression,

$$\sigma_{\text{low}} = \left(A + B\varepsilon_p^n\right)\left[1 + C_1 \ln\left(\frac{\dot{\varepsilon}_p}{\dot{\varepsilon}_0}\right)\right]\left(1 - T^{*m_{\text{th}}}\right), \tag{23}$$

whenever $\dot{\varepsilon}_p \le \dot{\varepsilon}_c^{\text{eff}}$. The high-rate branch is evaluated in two steps. First, the low-rate JC branch is continued as the baseline contribution,

$$\sigma_{\text{JC,low}} = \left(A + B\varepsilon_p^n\right)\left[1 + C_1 \ln\left(\frac{\dot{\varepsilon}_p}{\dot{\varepsilon}_0}\right)\right]\left(1 - T^{*m_{\text{th}}}\right), \tag{24}$$

and second, the drag increment is added according to Eq. (21). Thus the implemented high-rate branch is

$$\sigma_{\text{high}} = \sigma_{\text{JC,low}} + \sigma_{\text{drag}}, \tag{25}$$

which guarantees that the transition from the low-rate branch remains continuous because the logarithmic drag increment vanishes at the transition threshold.

This specific form reflects a conclusion reached repeatedly during model development: the high-rate branch should not be written as a drag-only stress. If the drag term alone is used above the transition, the total flow stress can drop unphysically at the switching point because the drag increment is constructed to start from zero. The implemented form therefore keeps the ordinary JC contribution as the baseline stress and adds only the incremental drag contribution.

3.2 Discrete Plastic-Strain-Rate and Adiabatic Heating Update

At the discrete level, the plastic strain rate is evaluated from the peridynamic kinematics and local plasticity update. This quantity is then used in three separate places within the constitutive evaluation:

1. the low-rate JC logarithmic rate term,
2. the logarithmic over-threshold drag function in the high-rate branch,
3. the adiabatic temperature update through the Taylor–Quinney relation.

The temperature update is written in incremental form as

$$\Delta T = \frac{\beta_{\text{TQ}}\,\sigma_{\text{flow}}\,\Delta\varepsilon_p}{\rho c_p}. \tag{26}$$

The updated temperature enters the constitutive law through the normalized temperature T^* and through the modulus ratio $G(T)/G_0$. This creates a direct thermomechanical coupling between plastic flow, heat generation, and drag-controlled strengthening.

The implementation also preserves the conceptual separation established in the constitutive derivation. The quantity c_{T0} is used only as the fixed velocity scale needed to define the microscopic transition-rate estimate, while $G(T)$ is used only as the temperature-dependent amplitude factor in the drag term. This avoids enforcing an artificial temperature dependence on the sound-speed scale inside the constitutive closure.

3.3 Calibration Strategy

The final constitutive law contains a clear separation between fixed material scales and calibrated closure parameters. The quantities b, c_{T0}, G_0, ρ, c_p, and the low-rate JC parameters are treated as material inputs. The bridge factor Λ_{ph} is set from the discretization-based area scaling in Eq. (17). The remaining drag parameters then play distinct roles:

- K_0 controls the overall magnitude of the drag increment,
- a controls how strongly the drag amplitude follows the thermal degradation of the shear modulus.

This separation was one of the key practical requirements identified during the development of the model: the drag branch must be simple enough to calibrate against high-rate impact data without obscuring the interpretation of the low-rate JC parameters.

4 Conclusion

A piecewise Johnson–Cook constitutive framework with a phonon-drag-motivated high-rate increment has been reconstructed from the model-development logic established in the present study. The final form preserves the conventional low-rate JC branch, introduces a fixed effective transition strain rate tied to the material scales c_{T0} and b through the bridge factor Λ_{ph}, and adds a logarithmic drag increment that is continuous at the transition.

Several modeling decisions were essential. First, the characteristic sound-speed scale was kept constant, while the temperature dependence was retained only in the shear-modulus factor $G(T)$. Second, the critical strain rate was treated as a material-dependent constitutive threshold rather than a direct function of impact velocity. Third, the high-rate branch was written as the continuing low-rate JC baseline plus a drag increment, rather than as a drag-only stress. This preserved continuity and avoided the nonphysical stress drop that appears when the drag term alone is used above the transition.

The resulting closure is not a full microscopic dislocation model, but it provides a compact and physically interpretable constitutive form for high-rate cold-spray impact simulations. It separates the roles of hardening, low-rate JC rate sensitivity, thermal softening, transition-rate bridging, and high-rate drag amplification in a way that can be implemented directly in the numerical solver and calibrated against impact data.

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